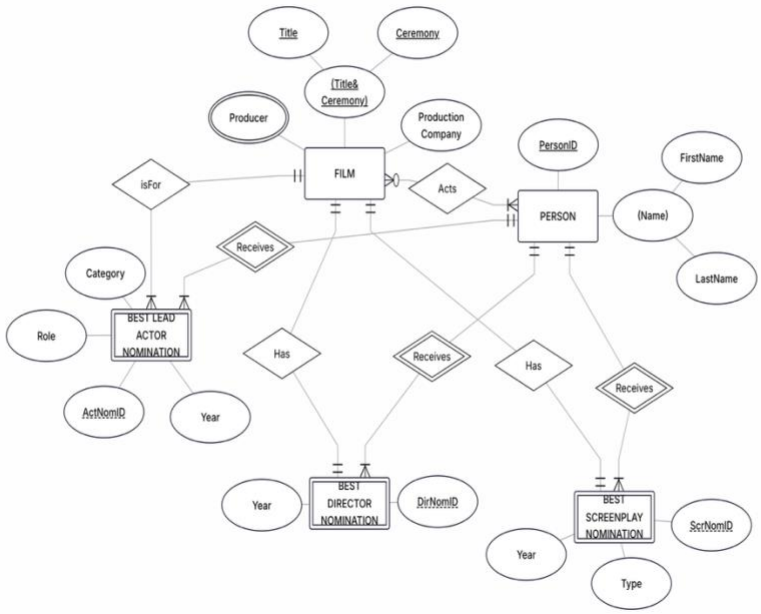
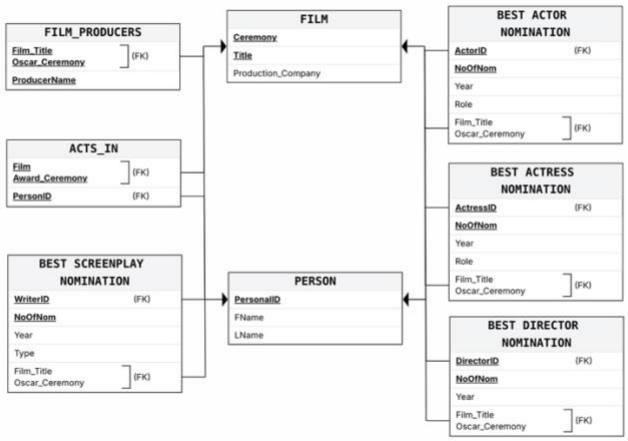
Project Proposal: The domain for which my database is based on would be the Big 5 Academy Award winners and nominees. This database would keep track of 5 entities which are Best Picture Nominee, Best Director Nominee, Best Actor Nominee, Best Actress Nominee, and Best Screenplay Nominee. This database and ER diagram will consist of films nominated for the "Big Five" awards at the academy awards which are Best Picture, Best Director, Best Actor, Best Actress, and Best Screenplay (Original or Adapted). The ER diagram would describe the relationship these nominations have with one another and accounting for a film and its actors, director(s), and writer(s) that are nominated in their respective category for each year this instance occurred.

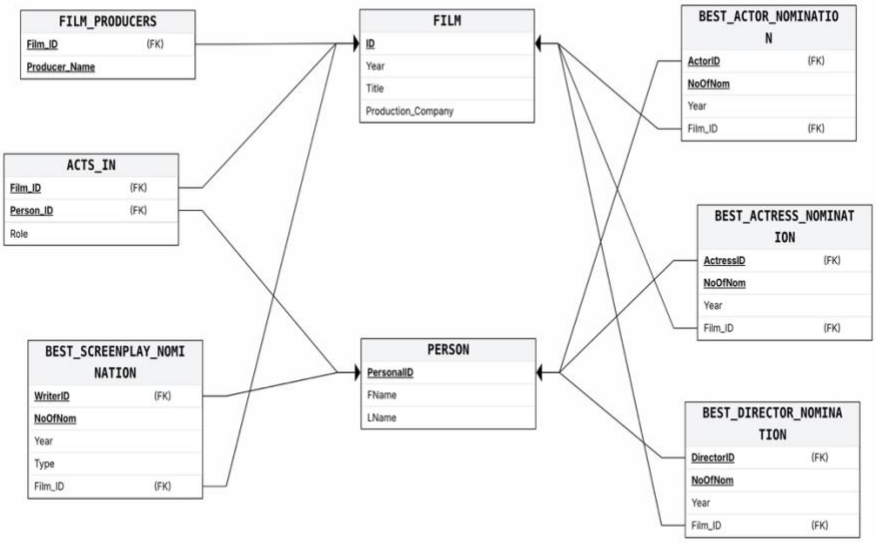
ERD DIAGRAM



RELATIONAL SCHEMA



RELATIONAL SCHEMA 3NF



TABLES WITH ENTITIES AND ATTRIBUTES

FILM

Title	**Ceremony**	Production Company

FILM_PRODUCERS

Film_Title (FK)	**Oscar_Ceremony** (FK)	**Producer_Name**

PERSON

PersonalID	First_Name	Last_Name

ACTS_IN

PersonID (FK)	**Film** (FK)	**Award_Ceremony** (FK)

BEST ACTOR NOMINATION

ActorID (FK)	**NoOfNom**	Year	Oscar_Ceremony (FK)	Film_Title (FK)	Role

BEST ACTRESS NOMINATION

ActressID (FK)	**NoOfNom**	Year	Oscar_Ceremony (FK)	Film_Title (FK)	Role

BEST DIRECTOR NOMINATION

DirectorID (FK)	**NoOfNom**	Year	Oscar_Ceremony (FK)	Film_Title (FK)

BEST SCREENPLAY NOMINATION

WriterID (FK)	**NoOfNom**	Year	Oscar_Ceremony (FK)	Film_Title (FK)	Type

CREATING SCHEMAS ON MYSQL

```
CREATE TABLE FILM (
  ID VARCHAR(4) NOT NULL,
  Ceremony VARCHAR(5) NOT NULL,
  Title VARCHAR(50) NOT NULL,
  Production_Company VARCHAR(50) NOT NULL,
  PRIMARY KEY (ID)
);

CREATE TABLE FILM_PRODUCERS (
  Film_ID VARCHAR(4) NOT NULL,
  Producer_Name VARCHAR(50) NOT NULL,
  PRIMARY KEY (Film_ID, Producer_Name),
  FOREIGN KEY (Film_ID) REFERENCES FILM(ID)
);

CREATE TABLE PERSON (
  Personal_ID CHAR(7) NOT NULL,
  First_Name VARCHAR(50) NOT NULL,
  Last_Name VARCHAR(50) NOT NULL,
  PRIMARY KEY (Personal_ID)
);
```

```
CREATE TABLE BEST_ACTOR_NOMINATION (
  NoOfNom INT NOT NULL,
  Actor_ID CHAR(7) NOT NULL,
  Year INT NOT NULL,
  Role VARCHAR(50) NOT NULL,
  Film_ID VARCHAR(4) NOT NULL,
  PRIMARY KEY (Actor_ID, NoOfNom),
  FOREIGN KEY (Actor_ID) REFERENCES PERSON(Personal_ID),
  FOREIGN KEY (Film_ID) REFERENCES FILM(ID)
);

CREATE TABLE BEST_ACTRESS_NOMINATION (
  NoOfNom INT NOT NULL,
  Actress_ID CHAR(7) NOT NULL,
  Year INT NOT NULL,
  Role VARCHAR(50) NOT NULL,
  Film_ID VARCHAR(4) NOT NULL,
  PRIMARY KEY (Actress_ID, NoOfNom),
  FOREIGN KEY (Actress_ID) REFERENCES PERSON(Personal_ID),
  FOREIGN KEY (Film_ID) REFERENCES FILM(ID)
);
```

```
CREATE TABLE BEST_DIRECTOR_NOMINATION (
  NoOfNom INT NOT NULL,
  Director_ID CHAR(7) NOT NULL,
  Year INT NOT NULL,
  Film_ID VARCHAR(4) NOT NULL,
  PRIMARY KEY (Director_ID, NoOfNom),
  FOREIGN KEY (Director_ID) REFERENCES PERSON(Personal_ID),
  FOREIGN KEY (Film_ID) REFERENCES FILM(ID)
);
CREATE TABLE BEST_SCREENPLAY_NOMINATION (
  NoOfNom INT NOT NULL,
  Year INT NOT NULL,
  Type VARCHAR(20) NOT NULL,
  Writer_ID CHAR(7) NOT NULL,
  Film_ID VARCHAR(4) NOT NULL,
  PRIMARY KEY (Writer_ID, NoOfNom),
  FOREIGN KEY (Writer_ID) REFERENCES PERSON(Personal_ID),
  FOREIGN KEY (Film_ID) REFERENCES FILM(ID)
);
CREATE TABLE ACTS_IN (
  Person_ID CHAR(7) NOT NULL,
  Film_ID VARCHAR(4) NOT NULL,
  PRIMARY KEY (Film_ID, Person_ID),
  FOREIGN KEY (Film_ID) REFERENCES FILM(ID),
  FOREIGN KEY (Person_ID) REFERENCES PERSON(Personal_ID)
);
```

INSERTING DATA

```
-- phpMyAdmin SQL Dump
-- version 6.0.0-dev+20241219.8e721911f0
-- https://www.phpmyadmin.net/
--
-- Host: 192.168.30.23
-- Generation Time: Oct 27, 2025 at 01:25 AM
-- Server version: 8.0.18
-- PHP Version: 8.2.26

SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO";
START TRANSACTION;
SET time_zone = "+00:00";



/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;


--
-- Database: `ProjectDatabase`
--
CREATE DATABASE IF NOT EXISTS `ProjectDatabase` DEFAULT CHARACTER SET utf8mb4
COLLATE utf8mb4_0900_ai_ci;
USE `ProjectDatabase`;
```

```
-- ------------------------------------------------------------

--
-- Table structure for table `ACTS_IN`
--

CREATE TABLE `ACTS_IN` (
  `Person_ID` char(7) NOT NULL,
  `Film_ID` varchar(4) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

--
-- Dumping data for table `ACTS_IN`
--

INSERT INTO `ACTS_IN` (`Person_ID`, `Film_ID`) VALUES
('2358016', 'A041'),
('4215399', 'A122'),
('4783529', 'A101'),
('5071398', 'A071'),
('5071398', 'A121'),
('7134980', 'A121'),
('7801954', 'A041'),
('8255001', 'A071'),
('8304761', 'A122'),
```

```
('9910238', 'A101');

-- ----------------------------------------------------------

--
-- Table structure for table `BEST_ACTOR_NOMINATION`
--

CREATE TABLE `BEST_ACTOR_NOMINATION` (
  `NoOfNom` int(11) NOT NULL,
  `Actor_ID` char(7) NOT NULL,
  `Year` int(11) NOT NULL,
  `Role` varchar(50) NOT NULL,
  `Film_ID` varchar(4) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

--
-- Dumping data for table `BEST_ACTOR_NOMINATION`
--

INSERT INTO `BEST_ACTOR_NOMINATION` (`NoOfNom`, `Actor_ID`, `Year`, `Role`,
`Film_ID`) VALUES
(1, '4783529', 1937, 'Norman Maine', 'A101'),
(1, '5071398', 1934, 'Peter Warne', 'A071'),
(2, '5071398', 1939, 'Rhett Butler', 'A121'),
(1, '7801954', 1931, 'Yancey Cravat', 'A041'),
```

```
(1, '8304761', 1939, 'Charles Chipping', 'A122');

-- ----------------------------------------------------------

--
-- Table structure for table `BEST_ACTRESS_NOMINATION`
--

CREATE TABLE `BEST_ACTRESS_NOMINATION` (
  `NoOfNom` int(11) NOT NULL,
  `Actress_ID` char(7) NOT NULL,
  `Year` int(11) NOT NULL,
  `Role` varchar(50) NOT NULL,
  `Film_ID` varchar(4) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

--
-- Dumping data for table `BEST_ACTRESS_NOMINATION`
--

INSERT INTO `BEST_ACTRESS_NOMINATION` (`NoOfNom`, `Actress_ID`, `Year`,
`Role`, `Film_ID`) VALUES
(1, '2358016', 1931, 'Sabra Cravat', 'A041'),
(1, '4215399', 1939, 'Katherine Ellis', 'A122'),
(1, '7134980', 1939, 'Scarlett O\'Hara', 'A121'),
(1, '8255001', 1934, 'Ellen Andrews', 'A071'),
```

```
(1, '9910238', 1937, 'Vicki Lester', 'A101');

-- ----------------------------------------------------------

--
-- Table structure for table `BEST_DIRECTOR_NOMINATION`
--

CREATE TABLE `BEST_DIRECTOR_NOMINATION` (
  `NoOfNom` int(11) NOT NULL,
  `Director_ID` char(7) NOT NULL,
  `Year` int(11) NOT NULL,
  `Film_ID` varchar(4) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

--
-- Dumping data for table `BEST_DIRECTOR_NOMINATION`
--

INSERT INTO `BEST_DIRECTOR_NOMINATION` (`NoOfNom`, `Director_ID`, `Year`,
`Film_ID`) VALUES
(1, '1492736', 1931, 'A041'),
(1, '3045671', 1939, 'A121'),
(1, '3608245', 1934, 'A071'),
(1, '5891203', 1939, 'A122'),
(1, '6429105', 1937, 'A101');
```

```
-- ----------------------------------------------------------

--
-- Table structure for table `BEST_SCREENPLAY_NOMINATION`
--

CREATE TABLE `BEST_SCREENPLAY_NOMINATION` (
  `NoOfNom` int(11) NOT NULL,
  `Year` int(11) NOT NULL,
  `Type` varchar(20) NOT NULL,
  `Writer_ID` char(7) NOT NULL,
  `Film_ID` varchar(4) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

--
-- Dumping data for table `BEST_SCREENPLAY_NOMINATION`
--

INSERT INTO `BEST_SCREENPLAY_NOMINATION` (`NoOfNom`, `Year`, `Type`,
`Writer_ID`, `Film_ID`) VALUES
(1, 1937, 'Original', '1057963', 'A101'),
(1, 1934, 'Adapted', '1093487', 'A071'),
(1, 1939, 'Adapted', '1932540', 'A122'),
(1, 1939, 'Adapted', '2560874', 'A121'),
(1, 1939, 'Adapted', '3704892', 'A122'),
```

```
(1, 1937, 'Original', '4981320', 'A101'),
(1, 1937, 'Original', '6429105', 'A101'),
(1, 1939, 'Adapted', '6678041', 'A122'),
(1, 1937, 'Original', '8123645', 'A101'),
(1, 1931, 'Adapted', '9174632', 'A041');

-- ----------------------------------------------------------

--
-- Table structure for table `FILM`
--

CREATE TABLE `FILM` (
  `ID` varchar(4) NOT NULL,
  `Year` int(11) NOT NULL,
  `Ceremony` varchar(5) NOT NULL,
  `Title` varchar(50) NOT NULL,
  `Production_Company` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

--
-- Dumping data for table `FILM`
--

INSERT INTO `FILM` (`ID`, `Year`, `Ceremony`, `Title`, `Production_Company`)
VALUES
```

```
('A041', 1931, '4th', 'Cimarron', 'RKO Radio Pictures'),
('A071', 1934, '7th', 'It Happened One Night', 'Columbia Pictures'),
('A101', 1937, '10th', 'A Star Is Born', 'Selznick International Pictures'),
('A121', 1939, '12th', 'Gone with the Wind', 'Metro-Goldwyn-Mayer'),
('A122', 1939, '12th', 'Goodbye, Mr. Chips', 'Metro-Goldwyn-Mayer');

-- ----------------------------------------------------------


--
-- Table structure for table `FILM_PRODUCERS`
--

CREATE TABLE `FILM_PRODUCERS` (
  `Film_ID` varchar(4) NOT NULL,
  `Producer_Name` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;


--
-- Dumping data for table `FILM_PRODUCERS`
--

INSERT INTO `FILM_PRODUCERS` (`Film_ID`, `Producer_Name`) VALUES
('A041', 'William LaBaron'),
('A071', 'Frank Capra'),
('A071', 'Harry Cohn'),
('A101', 'David O. Selznick'),
```

```
('A121', 'David O. Selznick'),
('A122', 'Victor Saville');

-- ----------------------------------------------------------

--
-- Table structure for table `PERSON`
--

CREATE TABLE `PERSON` (
  `Personal_ID` char(7) NOT NULL,
  `First_Name` varchar(50) NOT NULL,
  `Last_Name` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

--
-- Dumping data for table `PERSON`
--

INSERT INTO `PERSON` (`Personal_ID`, `First_Name`, `Last_Name`) VALUES
('1057963', 'Dorothy', 'Parker'),
('1093487', 'Robert', 'Riskin'),
('1492736', 'Wesley', 'Ruggles'),
('1932540', 'Claudine', 'West'),
('2358016', 'Irene', 'Dunne'),
('2560874', 'Sidney', 'Howard'),
```

```
('3045671', 'Victor', 'Fleming'),
('3608245', 'Frank', 'Capra'),
('3704892', 'Eric', 'Maschwitz'),
('4215399', 'Greer', 'Garson'),
('4783529', 'Fredric', 'March'),
('4981320', 'Robert', 'Carson'),
('5071398', 'Clark', 'Gable'),
('5891203', 'Sam', 'Wood'),
('6429105', 'William', 'Wellman'),
('6678041', 'R.C', 'Sherriff'),
('7134980', 'Vivien', 'Leigh'),
('7801954', 'Richard', 'Dix'),
('8123645', 'Allan', 'Campbell'),
('8255001', 'Claudette', 'Colbert'),
('8304761', 'Robert', 'Donat'),
('9174632', 'Howard', 'Estabrook'),
('9910238', 'Janet', 'Gaynor');

COMMIT;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
```

PROJECT QUERIES

1. The purpose of this query is that it showcases the total number of films for each year in the database. The COUNT aggregate function is used to count the number a times a year appears in the relation film. The GROUP BY clause filters by year. For example, 1931, 1934, and 1937 each had one film that was nominated for all the big five categories at the Academy Awards and 1939 had 2 films that achieved this feat in the same year. This query is useful for checking the statistics on this feat happening either once or multiple times each year, for data purposes, and so one does not have to manually count the number of films for each year in the database. Also, the ORDER BY orders the relation by the highest number of total films each year.

```
SELECT Year, COUNT(year) AS numOfFilms

FROM film

GROUP BY year

ORDER BY numOfFilms DESC
```

Year	NumOfFilms
1939	2
1931	1
1934	1
1937	1

2. This query showcases the full name of an actor and the number of times that actor was nominated for Best Actor in the database. The query uses the relations persons and best_actor_nomination to configure this query. The GROUP BY and ORDER BY clauses are used to filter the table by actor using the actor_id and order the table by the highest number of total Best Actor nominations respectively. The results mean that of the current actors in the database Clark Gable has the most Best Actors nominations with 2. The results might be useful for again statistic wise checking which actor has the most Acting nominations in the database that is specifically for films that achieved all big five nominations at the Academy awards. Also, this specific query I very useful because with a few tweaks to the FROM and WHERE clauses you can get the results for the number of times an actress is nominated for Best Actress in the database as well for the writers and directors.

SELECT BA.actor_id,p.first_name,p.last_name,COUNT(noofnom) AS totalnominations

FROM persons AS P,best_actor_nomination AS BA

WHERE P.personal_id = BA.actor_id

GROUP BY BA.actor_id

Order by totalnominations DESC

Actor_id	First_name	Last_name	totalNominations
5071398	Clark	Gable	2
4783529	Fredric	March	1
7801954	Richard	Dix	1
8304761	Robert	Donat	1

3. This query showcases a film's title, year, and ceremony along with the actress nominated for Best Actress in the movie. The query uses the three relations: best_actress_nomination persons, and film. The results mean that Irene Dunne was nominated for her performance in Cimarron, Claudette Colbert was nominated for It Happened One Night, and so on. The ORDER BY clause orders the relation by year. These results are useful since it identifies the film by name that an actress was nominated instead of a film_id like in the best_actress_nomination relation. This query provides more information about the films and performances nominated. Also, like the query above this query is very useful because with a few tweaks to the FROM and WHERE clauses you can get the same type of results but for Best Actor.

SELECT F.year,F.ceremony,F.title,P.first_name,P.last_name

FROM best_actress_nomination AS BA,film AS F,persons as P

WHERE BA.film_id = F.id AND BA.actress_id = P.personal_id

Order by f.year

Year	Ceremony	Title	First_name	Last_name
1931	4th	Cimarron	Irene	Dunne
1934	7th	It Happened One Night	Claudette	Colbert
1937	10th	A Star Is Born	Janet	Gaynor
1939	12th	Gone with the Wind	Vivien	Leigh

1939	12th	Goodbye, Mr. Chips	Greer	Garson

4. This query showcases the number of writers a film has and the whether the films screenplay is original or an adaption. The query uses two relations: film and best_screenplay_nomination. The COUNT aggregate function is used to count the number of writers using the writer_id from the best_screenplay_nomination relation. The ORDER BY clause is used to order the relation by number of writers form greatest to least. The results mean that of the films in the database four of the screenplays are Adapted and one is Original. As well as A Star Is Born having four writers the most in the database and several films having only one writer. These results are useful because we get more information about a film and its screenplay nomination without having to search through multiple relations.

```
SELECT F.ID, F.year, F.ceremony, F.title,COUNT(BS.writer_id) AS numofwriters, BS.type as screenplaytype

FROM film AS F, best_screenplay_nomination AS BS

WHERE F.id = BS.film_id

GROUP BY F.id

ORDER BY numofwriters DESC
```

ID	Year	Ceremony	Title	NumOfWriters	ScreenplayType
A101	1937	10TH	A Star Is Born	4	Original
A122	1939	12TH	Goodbye, Mr. Chips	3	Adapted
A041	1931	4TH	Cimarron	1	Adapted
A071	1934	7TH	It Happened One Night	1	Adapted
A121	1939	12th	Gone with the Wind	1	Adapted

5. This query showcases a person nominated for Best Director and Best Screenplay for the same film. The query uses four relations: persons, best_director_nomiantion,

best_screenplay_nomination, and film. The results of this query mean that of the current people in the database William Wellman is only person to be nominated for both best director and screenplay for the same film which was A Star Is Born in 1937. The query and results are useful because we can identify something relevant in the database as directors often do write the screenplay for their movie. However, in this case there is only one instance of this. Also, again you can tweak FROM and WHERE clause to determine a person nominated for both Best Actor and Director or Best Actress and Director.

```
SELECT F.year,BD.director_id,P.first_name,P.last_name,F.title

FROM best_director_nomination AS BD,best_screenplay_nomination AS BS,persons AS P,film as F

WHERE BD.director_id = BS.writer_id AND BD.film_id = BS.film_id AND BD.director_id = P.personal_id AND F.id = BD.film_id

GROUP BY BD.director_id
```

Year	Director_Id	First_name	Last_Name	Title
1937	6429105	William	Wellman	A Star Is Born